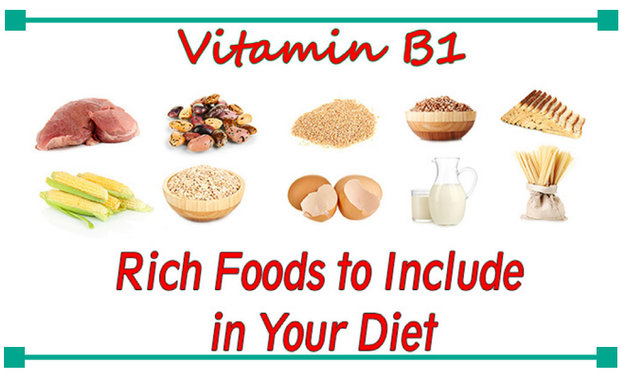
- **Deficiency:** It manifests in the form of defective blood clotting. Low levels of prothrombin and hemorrhage are seen in severe forms of deficiency.

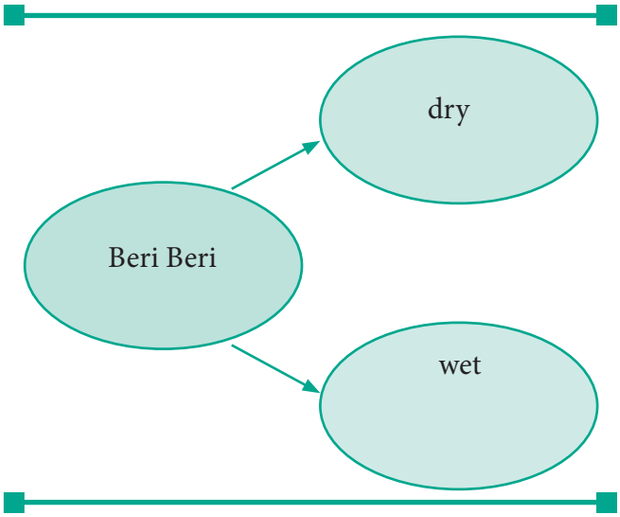
5.3.2.2 Water Soluble Vitamins

(a) Thiamin (B1): Thiamin is the first member of the B complex vitamins which is essential to the body in its coenzyme form.

- **Functions:** The coenzyme of thiamin is Thiamin Pyro Phosphate (TPP). Thiamin is useful in our body for the following functions
  - It enhances growth in human beings.
  - It plays an important role as a coenzyme in carbohydrate metabolism
  - Maintenance of nerves in normal condition.
- **Food sources:** Good food sources include lean pork, beef, liver, whole or enriched grams and legumes.

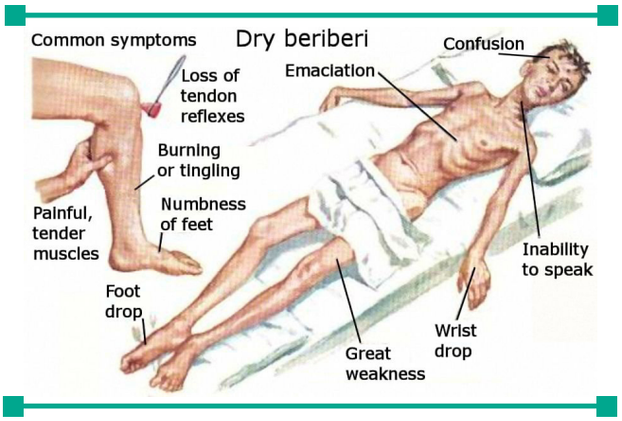


- **Deficiency of thiamin:** The discovery of thiamin provided the answer to the puzzle of a nutritional problem called beri beri. The deficiency of thiamin causes beri beri in human beings. The Philippino word beri beri means "I Can't" refers to the lack of neuromotor coordination in persons with the disease.



▲ Fig. 3 Types of Beri Beri

Beri Beri is of two types: dry and wet type. In dry beri beri the muscles become progressively wasted, weak and walking becomes difficult. If not treated the patient becomes bedridden and will die. In wet beri beri edema is present which involves the face, trunk and serous cavities. Palpitation and breathlessness are present. The heart becomes weak



Chapter5_088-119.indd 19 18-01-2019 12:29:17